

Decent Coaching Classes
X SB
Worksheet

Maths Part 1 (Chapters 1 to 6) | Maths Part 2 (Chapters 1, 2, 5, 6, 7) | Maharashtra State Board
Show all steps of your working. Draw neat figures wherever required. Take $\pi = 22/7$ unless stated otherwise.

MATHS PART 1 (ALGEBRA)

Part 1 – Chapter 1 : Linear Equations in Two Variables

Q1. Choose the correct alternative.

- (1) Which of the following is a linear equation in two variables?
(A) $3x^2 - 7y = 13$ (B) $4xy - 5y - 8 = 0$ (C) $4m + 3n = 12$ (D) $4/x + 5/y = 4$
- (2) Find the value of the determinant:
$$\begin{vmatrix} 5 & 3 \\ -2 & 4 \end{vmatrix}$$

(A) 14 (B) 26 (C) 20 (D) -26
- (3) For the equations $3x + 2y = 7$ and $x - y = 1$, the value of D is
(A) -5 (B) 5 (C) 1 (D) -1
- (4) The solution of $x + y = 5$ and $x - y = 1$ is
(A) (2, 3) (B) (3, 2) (C) (4, 1) (D) (1, 4)
- (5) The graph of the equation $x = 4$ is a line parallel to the
(A) X-axis (B) Y-axis (C) line $y = x$ (D) none of these

Q2. Solve the following simultaneous equations.

- (1) $3a + 5b = 26$; $a + 5b = 22$
(2) $x + 7y = 10$; $3x - 2y = 7$
(3) $2x - 3y = 9$; $2x + y = 13$
(4) $5m - 3n = 19$; $m - 6n = -7$
(5) $99x + 101y = 499$; $101x + 99y = 501$
(6) $2/x + 3/y = 13$; $5/x - 4/y = -2$

Q3. Solve the following simultaneous equations using Cramer's rule.

- (1) $3x - 4y = 10$; $4x + 3y = 5$
(2) $x + 2y = -1$; $2x - 3y = 12$
(3) $3x + 5y = 17$; $2x - y = 7$

Q4. Solve the following simultaneous equations graphically.

- (1) $x + y = 6$; $x - y = 2$
(2) $2x + y = 8$; $x - y = 1$

Q5. Solve the following word problems.

- (1) The sum of two numbers is 63 and their difference is 17. Find the numbers.
- (2) The sum of the present ages of a father and his son is 50 years. Five years ago, the father's age was 9 times the son's age. Find their present ages.
- (3) A taxi charges a fixed amount plus a fixed rate per km. The fare for 10 km is ₹ 105 and for 15 km is ₹ 155. Find the fixed charge and the charge per km.
- (4) The sum of the digits of a two-digit number is 9. The number obtained by interchanging the digits exceeds the original number by 27. Find the original number.

Part 1 – Chapter 2 : Quadratic Equations

Q1. Choose the correct alternative.

- (1) Which of the following is a quadratic equation?
(A) $3x + 4 = 0$ (B) $x^3 - 2x = 5$ (C) $2x^2 - 7x = 0$ (D) $5/x = 3$
- (2) The roots of $x^2 - 5x + 6 = 0$ are
(A) 2 and 3 (B) -2 and -3 (C) 1 and 6 (D) -1 and -6
- (3) The discriminant of $x^2 + 4x + 4 = 0$ is
(A) 16 (B) 0 (C) 8 (D) -16
- (4) If the discriminant of a quadratic equation is negative, then its roots are
(A) real and equal (B) real and unequal (C) not real (D) rational
- (5) The sum of the roots of $2x^2 - 5x + 3 = 0$ is
(A) $5/2$ (B) $-5/2$ (C) $3/2$ (D) $2/3$
- (6) The quadratic equation whose roots are 3 and -2 is
(A) $x^2 - x - 6 = 0$ (B) $x^2 + x - 6 = 0$ (C) $x^2 - 5x + 6 = 0$ (D) $x^2 + x + 6 = 0$

Q2. Write the following equations in the standard form $ax^2 + bx + c = 0$ and state the values of a, b and c.

- (1) $2x^2 = 5x - 3$
- (2) $(x + 2)^2 = 3x + 7$

Q3. Solve the following quadratic equations by factorisation.

- (1) $x^2 - 7x + 12 = 0$
- (2) $2x^2 + 5x - 12 = 0$
- (3) $6m^2 - 13m + 6 = 0$

Q4. Solve by completing the square : $x^2 + 6x - 7 = 0$.

Q5. Solve using the quadratic formula.

- (1) $x^2 - 6x + 4 = 0$
- (2) $2y^2 - 7y + 3 = 0$

Q6. Nature of roots.

- (1) Find the discriminant and state the nature of the roots of : (a) $x^2 - 4x + 4 = 0$ (b) $2x^2 + 3x + 5 = 0$ (c) $x^2 - 3x - 1 = 0$
- (2) Find the value of k for which $x^2 + kx + 9 = 0$ has real and equal roots.

Q7. Relation between roots and coefficients.

- (1) Without solving, find the sum and the product of the roots of $3x^2 - 10x + 8 = 0$.
- (2) Form a quadratic equation whose roots are 4 and -7.
- (3) If α and β are the roots of $x^2 - 5x + 6 = 0$, find the value of $\alpha^2 + \beta^2$.

Q8. Solve the following word problems.

- (1) The product of two consecutive positive integers is 132. Find the integers.
- (2) The length of a rectangle is 3 m more than its breadth and its area is 70 m^2 . Find the length and the breadth.
- (3) The sum of the squares of two consecutive odd natural numbers is 130. Find the numbers.

Part 1 – Chapter 3 : Arithmetic Progression

Q1. Choose the correct alternative.

- (1) Which of the following sequences is an A.P.?
(A) 2, 4, 8, 16, ... (B) 3, 6, 9, 12, ... (C) 1, 4, 9, 16, ... (D) 1, 3, 6, 10, ...
- (2) The common difference of the A.P. 7, 3, -1, -5, ... is
(A) 4 (B) -4 (C) 3 (D) -3
- (3) The n th term of an A.P. with first term a and common difference d is
(A) $a + nd$ (B) $a + (n - 1)d$ (C) $a - (n - 1)d$ (D) $nd - a$
- (4) The 10th term of the A.P. 5, 8, 11, ... is
(A) 30 (B) 32 (C) 35 (D) 38
- (5) The sum of the first 10 natural numbers is
(A) 45 (B) 50 (C) 55 (D) 60

Q2. Write the first four terms of the A.P. whose first term a and common difference d are given.

- (1) $a = 10, d = 5$
- (2) $a = -3, d = -2$

Q3. Solve.

- (1) Find the 15th term of the A.P. 3, 7, 11, ...
- (2) Which term of the A.P. 4, 9, 14, ... is 109?
- (3) Is 301 a term of the A.P. 5, 11, 17, 23, ... ? Give reason.
- (4) In an A.P., $t_4 = 22$ and $t_9 = 47$. Find a, d and t_{15} .

Q4. Find the sum.

- (1) Sum of the first 20 terms of the A.P. 3, 7, 11, ...
- (2) $1 + 2 + 3 + \dots + 100$
- (3) The sum of all odd numbers from 1 to 100.
- (4) How many terms of the A.P. 24, 21, 18, ... must be taken so that the sum is 78?

Q5. Solve the following word problems.

- (1) A person saves ₹ 1000 in the first month and increases the saving by ₹ 200 every month. Find the total savings in 12 months.
- (2) In a hall, the first row has 20 chairs and every next row has 2 chairs more than the previous row. If there are 15 rows, find the total number of chairs.

Part 1 – Chapter 4 : Financial Planning

Q1. Choose the correct alternative.

- (1) GST stands for
(A) Goods and Sales Tax (B) Goods and Services Tax (C) General Sales Tax
(D) Government Service Tax
- (2) In an intra-state transaction, GST is divided equally into
(A) CGST and IGST (B) SGST and IGST (C) CGST and SGST (D) VAT and CGST
- (3) A share has face value ₹ 10 and market value ₹ 25. The share is at a premium of
(A) ₹ 10 (B) ₹ 15 (C) ₹ 25 (D) ₹ 35
- (4) $\text{NAV of one unit} \times \text{number of units} =$
(A) dividend (B) brokerage (C) total fund value (D) face value
- (5) The rate of GST on brokerage is
(A) 5% (B) 12% (C) 18% (D) 28%

Q2. GST.

- (1) A shopkeeper in Maharashtra sells goods of taxable value ₹ 8000 to a customer in the same state. The rate of GST is 12%. Find the CGST, the SGST and the total invoice value.
- (2) A manufacturer sells goods of taxable value ₹ 50,000 at 18% GST. He had bought raw material worth ₹ 30,000 on which he paid 18% GST. Find the input tax credit (ITC) and the GST payable by him. Also find the CGST and SGST payable.

Q3. Shares.

- (1) The face value of a share is ₹ 10 and its market value is ₹ 150. Find the premium.
- (2) The market value of a share is ₹ 200. Brokerage is 0.3%. Find the purchase value of one share.
- (3) Rohit sold 500 shares of market value ₹ 40 each. Brokerage is 0.5%. Find the amount he received.
- (4) Meena bought 200 shares of face value ₹ 100 and market value ₹ 120. The company declared a dividend of 15%. Find the total dividend and the rate of return on the investment (ignore brokerage).
- (5) Amit purchased 200 shares of market value ₹ 50. Brokerage is 0.5% and GST on brokerage is 18%. Find the brokerage, CGST, SGST and the total amount paid.
- (6) Sunil invested ₹ 40,200 in shares of market value ₹ 40 with brokerage 0.5% (ignore GST). How many shares did he buy? Find the total brokerage.

Q4. Mutual Fund and SIP.

- (1) The total value of a mutual fund scheme is ₹ 120 crore and 4 crore units are issued. Find the NAV of one unit.
- (2) The NAV of a unit is ₹ 25. How many units will be allotted for an investment of ₹ 15,000?
- (3) Kavita invests ₹ 2000 every month through SIP for 5 years. Find her total investment.

Part 1 – Chapter 5 : Probability**Q1. Choose the correct alternative.**

- (1) The probability of a sure event is
(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) -1
- (2) The probability of an event always lies between
(A) -1 and 1 (B) 0 and 1 (C) 0 and 100 (D) 1 and 2
- (3) When two coins are tossed, the number of outcomes in the sample space is
(A) 2 (B) 3 (C) 4 (D) 8
- (4) A die is rolled once. The probability that a prime number appears is
(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{6}$
- (5) If $P(A) = \frac{3}{7}$, then $P(A') =$
(A) $\frac{3}{7}$ (B) $\frac{4}{7}$ (C) $\frac{7}{3}$ (D) 1

Q2. Write the sample space and the number of outcomes.

- (1) Two coins are tossed simultaneously.
- (2) A die is rolled and a coin is tossed.

Q3. A die is rolled once. Find the probability of getting

- (1) an even number
- (2) a number greater than 4
- (3) a number less than 7
- (4) the number 7

Q4. Two dice are rolled simultaneously. Find the probability that

- (1) the sum of the numbers is 9
- (2) the sum of the numbers is 7
- (3) the same number appears on both dice

Q5. One card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is

- (1) a king
- (2) a red card
- (3) a face card
- (4) the ace of spades
- (5) a heart

Q6. A bag contains 5 red, 4 blue and 3 green balls. One ball is drawn at random. Find the probability that it is

- (1) red
- (2) not green
- (3) blue or green

Q7. Three coins are tossed simultaneously. Find the probability of getting

- (1) exactly two heads
- (2) at least one head

Q8. A number is selected at random from the numbers 1 to 25. Find the probability that the number is

- (1) a multiple of 5
- (2) a prime number

Part 1 – Chapter 6 : Statistics

Q1. Choose the correct alternative.

- (1) The class mark of the class 20 – 30 is
(A) 20 (B) 25 (C) 30 (D) 10
- (2) The modal class of a frequency distribution is the class having
(A) the highest class mark (B) the highest frequency (C) the lowest frequency
(D) the largest class width
- (3) In the assumed mean method, the mean is given by
(A) $A + \frac{\sum fd}{\sum f}$ (B) $A - \frac{\sum fd}{\sum f}$ (C) $\frac{\sum fd}{\sum f}$ (D) $A \times \frac{\sum fd}{\sum f}$
- (4) In a pie diagram, the central angle of a component is
(A) $(\text{value} / \text{total}) \times 100^\circ$ (B) $(\text{value} / \text{total}) \times 180^\circ$ (C) $(\text{value} / \text{total}) \times 360^\circ$ (D)
 $(\text{total} / \text{value}) \times 360^\circ$

Q2. Find the mean of the following data by the direct method and by the assumed mean method (take A = 25).

Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	5	8	15	16	6

Q3. Find the mean by the step deviation method (take A = 45).

Class	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
Frequency	4	8	14	10	4

Q4. Find the median of the following data.

Class	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency	4	11	20	9	6

Q5. Find the mode of the following data.

Class	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency	6	8	20	12	4

Q6. The monthly expenditure of a family is ₹ 24,000. It is spent as shown below. Find the central angle for each item to draw a pie diagram.

Item	Food	Rent	Education	Transport	Savings
Amount (₹)	8000	6000	4000	2000	4000

Q7. Draw a histogram and a frequency polygon for the data given in Q4.

MATHS PART 2 (GEOMETRY)

Part 2 – Chapter 1 : Similarity

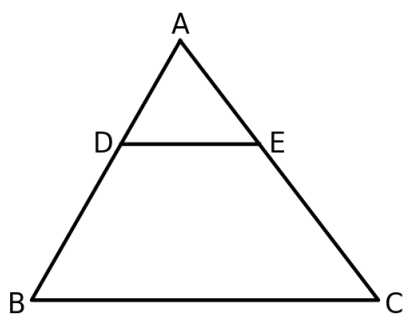


Figure 1 : In $\triangle ABC$, D lies on AB and E lies on AC.

Q1. Choose the correct alternative.

- (1) The ratio of the areas of two triangles having equal bases is equal to the ratio of their
(A) sides (B) heights (C) perimeters (D) angles
- (2) If $\triangle ABC \sim \triangle PQR$ and $AB/PQ = 2/3$, then $A(\triangle ABC) : A(\triangle PQR) =$
(A) 2 : 3 (B) 4 : 9 (C) 3 : 2 (D) 9 : 4
- (3) The ratio of the areas of two similar triangles is equal to the ratio of the
(A) corresponding sides (B) squares of the corresponding sides (C) perimeters
(D) corresponding angles
- (4) In $\triangle ABC$, ray AD bisects $\angle A$ and D lies on BC. Then
(A) $AB/AC = BD/DC$ (B) $AB/AC = DC/BD$ (C) $AB/BD = DC/AC$ (D) $AB \cdot AC = BD \cdot DC$

Q2. The base of $\triangle ABC$ is 9 cm and its height is 5 cm. The base of $\triangle PQR$ is 10 cm and its height is 6 cm. Find $A(\triangle ABC) : A(\triangle PQR)$.

Q3. In Figure 1, seg DE $\parallel$ seg BC.

- (1) If $AD = 4$, $DB = 6$ and $AE = 3$, find EC.
- (2) If $AD : DB = 2 : 3$ and $AC = 10$, find AE.

Q4. In $\triangle LMN$, points P and Q lie on LM and LN respectively. $LP = 3$, $PM = 6$, $LQ = 2$ and $QN = 4$. Is seg PQ $\parallel$ seg MN? Give reason.

Q5. In $\triangle ABC$, ray AD bisects $\angle A$ and D lies on side BC. If $AB = 6$, $AC = 9$ and $BD = 4$, find DC.

Q6. $\triangle ABC \sim \triangle PQR$. $A(\triangle ABC) = 81 \text{ cm}^2$ and $A(\triangle PQR) = 144 \text{ cm}^2$. If $PQ = 8 \text{ cm}$, find AB.

Q7. Using a test of similarity, decide whether the triangles are similar. Name the test.

- (1) In $\triangle ABC$ and $\triangle PQR$, $AB = 4$, $AC = 6$, $PQ = 6$, $PR = 9$ and $\angle A = \angle P$.
- (2) $\triangle XYZ$ has sides 3 cm, 4 cm, 5 cm and $\triangle LMN$ has sides 6 cm, 8 cm, 10 cm.

Q8. Three parallel lines are cut by two transversals. On the first transversal the intercepts are $AB = 6$ and $BC = 9$. On the second transversal the corresponding intercept $DE = 8$. Find EF .

Q9. A vertical pole 6 m high casts a shadow 4 m long. At the same time a tree casts a shadow 20 m long. Find the height of the tree.

Part 2 – Chapter 2 : Pythagoras Theorem

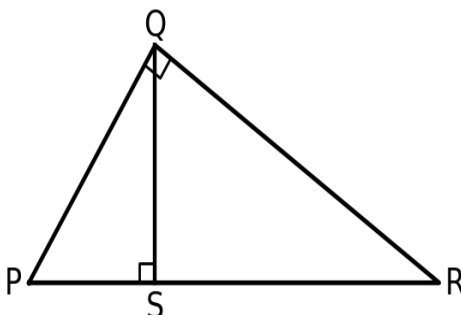


Figure 2 : ΔPQR is right-angled at Q and $\text{seg } QS \perp \text{seg } PR$.

Q1. Choose the correct alternative.

- (1) Which of the following is a Pythagorean triplet?
(A) (4, 5, 6) (B) (5, 12, 13) (C) (6, 7, 8) (D) (7, 8, 9)
- (2) In a right-angled triangle, the altitude on the hypotenuse divides it into parts p and q . The length of the altitude is
(A) $p + q$ (B) $p \times q$ (C) $\sqrt{p \times q}$ (D) $(p + q)/2$
- (3) The length of the diagonal of a square of side 5 cm is
(A) 5 cm (B) 10 cm (C) $5\sqrt{2}$ cm (D) 25 cm
- (4) Apollonius theorem : if M is the midpoint of BC in ΔABC , then $AB^2 + AC^2 =$
(A) $AM^2 + BM^2$ (B) $2AM^2 + 2BM^2$ (C) $2AM^2 + BM^2$ (D) $AM^2 + 2BM^2$

Q2. Find the unknown side of the right-angled triangle.

- (1) The two perpendicular sides are 9 cm and 12 cm. Find the hypotenuse.
- (2) The hypotenuse is 25 cm and one side is 7 cm. Find the other side.

Q3. Decide whether the triangle with the given sides is a right-angled triangle.

- (1) 7 cm, 24 cm, 25 cm
- (2) 5 cm, 6 cm, 8 cm

Q4. In Figure 2, $PS = 4$ and $SR = 9$. Find QS , PQ and QR .

Q5. In ΔABC , M is the midpoint of BC . $AB = 9$, $AC = 13$ and $AM = 10$. Find BC .

Q6. Solve.

- (1) A 10 m long ladder rests against a vertical wall. Its top reaches 8 m above the ground. How far is the foot of the ladder from the wall?
- (2) The diagonal of a rectangle is 41 cm and one side is 40 cm. Find the other side.
- (3) Find the altitude of an equilateral triangle of side 12 cm.
- (4) The diagonals of a rhombus are 16 cm and 30 cm. Find its side.

Part 2 – Chapter 5 : Co-ordinate Geometry

Q1. Choose the correct alternative.

- (1) The distance of the point (3, 4) from the origin is
(A) 3 (B) 4 (C) 5 (D) 7

- (2) The midpoint of the segment joining (2, 4) and (6, 8) is
 (A) (4, 6) (B) (8, 12) (C) (2, 2) (D) (3, 4)
- (3) The slope of the line passing through (1, 2) and (3, 6) is
 (A) 1 (B) 2 (C) 3 (D) 4
- (4) The slope of the X-axis is
 (A) 0 (B) 1 (C) -1 (D) not defined
- (5) The distance between the points (-2, 3) and (4, -5) is
 (A) 6 (B) 8 (C) 10 (D) 14

Q2. Find the distance between the following pairs of points.

- (1) A(-1, 7) and B(4, -5)
 (2) P(0, 0) and Q(-6, 8)

Q3. Show that the points A(0, 0), B(3, 0) and C(3, 4) are the vertices of a right-angled triangle.

Q4. Show that the points A(1, 1), B(3, 3) and C(5, 5) are collinear.

Q5. Midpoint.

- (1) Find the midpoint of the segment joining (-4, 7) and (6, -3).
 (2) The midpoint of seg AB is (3, 5) and A is (1, 2). Find the coordinates of B.

Q6. Section formula.

- (1) Find the coordinates of the point which divides the segment joining A(1, 7) and B(6, -3) internally in the ratio 2 : 3.
 (2) Find the coordinates of the centroid of the triangle with vertices (1, 2), (3, -4) and (5, 8).

Q7. Slope of a line.

- (1) Find the slope of the line passing through (-2, 5) and (4, -1).
 (2) Find the slope of a line which makes an angle of 45° with the positive direction of the X-axis. What is the slope if the angle is 60° ?
 (3) Show that the line through (1, 2) and (3, 6) is parallel to the line through (0, -1) and (2, 3).

Q8. Find x, if the distance between P(x, 2) and Q(3, -2) is 5.

Part 2 – Chapter 6 : Trigonometry

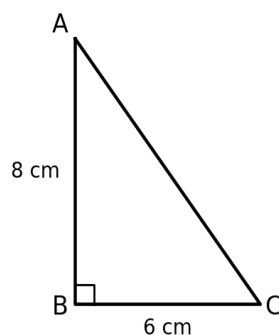


Figure 3 : $\triangle ABC$, $\angle B = 90^\circ$.

Q1. Choose the correct alternative.

- (1) $\sin^2\theta + \cos^2\theta =$
 (A) 0 (B) 1 (C) 2 (D) $\tan^2\theta$
- (2) $1 + \tan^2\theta =$
 (A) $\cot^2\theta$ (B) $\sec^2\theta$ (C) $\operatorname{cosec}^2\theta$ (D) $\sin^2\theta$

- (3) The value of $\tan 45^\circ$ is
 (A) 0 (B) $1/2$ (C) 1 (D) $\sqrt{3}$
- (4) If $\sin\theta = 3/5$, then $\cos\theta =$
 (A) $4/5$ (B) $3/4$ (C) $5/4$ (D) $5/3$
- (5) The value of $\sin 30^\circ + \cos 60^\circ$ is
 (A) $1/2$ (B) 1 (C) $\sqrt{3}/2$ (D) 0

Q2. In Figure 3, $AB = 8$ cm and $BC = 6$ cm. Find AC and then write $\sin A$, $\cos A$ and $\tan A$.

Q3. Evaluate.

- (1) $\sin 30^\circ \times \cos 60^\circ + \tan 45^\circ$
 (2) $2 \sin^2 45^\circ + 3 \cos^2 60^\circ$
 (3) $\sin 60^\circ \times \cos 30^\circ + \cos 60^\circ \times \sin 30^\circ$

Q4. If $\sin\theta = 5/13$, find $\cos\theta$ and $\tan\theta$.

Q5. If $\sec\theta = 13/12$, find the values of the other five trigonometric ratios.

Q6. Prove the following identities.

- (1) $(1 - \sin^2\theta)(1 + \tan^2\theta) = 1$
 (2) $\sin\theta / (1 + \cos\theta) + (1 + \cos\theta) / \sin\theta = 2 \operatorname{cosec}\theta$
 (3) $(\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta) = 1$

Q7. Heights and distances (take $\sqrt{3} = 1.73$).

- (1) A 10 m long ladder leans against a wall and makes an angle of 60° with the ground. How high on the wall does it reach?
- (2) The shadow of a tower is 30 m long when the angle of elevation of the sun is 45° . Find the height of the tower.
- (3) From the top of a 60 m high building, the angle of depression of a car on the ground is 30° . Find the distance of the car from the foot of the building.
- (4) From a point on the ground 20 m away from the foot of a tower, the angle of elevation of the top of the tower is 60° . Find the height of the tower.

Part 2 – Chapter 7 : Mensuration

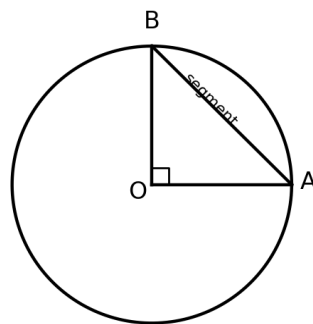


Figure 4 : Circle with centre O and radius 10 cm, $\angle AOB = 90^\circ$.

Q1. Choose the correct alternative.

- (1) The length of an arc of a circle of radius r and central angle θ is
 (A) $(\theta/360) \times \pi r^2$ (B) $(\theta/360) \times 2\pi r$ (C) $(\theta/180) \times \pi r^2$ (D) $\theta \times 2\pi r$
- (2) The area of a sector of a circle of radius r and central angle θ is
 (A) $(\theta/360) \times \pi r^2$ (B) $(\theta/360) \times 2\pi r$ (C) $(360/\theta) \times \pi r^2$ (D) πr^2
- (3) The total surface area of a solid hemisphere of radius r is
 (A) $2\pi r^2$ (B) $3\pi r^2$ (C) $4\pi r^2$ (D) πr^2

- (4) The volume of a cone of radius r and height h is
(A) $\pi r^2 h$ (B) $(1/3)\pi r^2 h$ (C) $(2/3)\pi r^3$ (D) $\pi r l$

Q2. Find the length of the arc and the area of the sector.

- (1) Radius = 14 cm, central angle = 90°
(2) Radius = 21 cm, central angle = 60°

Q3. In Figure 4, find the area of the minor segment (take $\pi = 3.14$).

Q4. Find the required surface area and volume.

- (1) A cylinder of radius 7 cm and height 10 cm : curved surface area and volume.
(2) A cone of radius 6 cm and height 8 cm ($\pi = 3.14$) : slant height, total surface area and volume.
(3) A sphere of radius 21 cm : surface area and volume.
(4) A solid hemisphere of radius 7 cm : curved surface area, total surface area and volume.

Q5. Solve.

- (1) A solid metal sphere of radius 6 cm is melted and recast into a cone of height 24 cm. Find the radius of the cone.
(2) A cylindrical water tank has radius 1.4 m and height 3 m. How many litres of water can it hold? ($1 \text{ m}^3 = 1000 \text{ litres}$)
(3) A toy is in the shape of a cone of height 4 cm mounted on a hemisphere of radius 3 cm (same base). Find the volume of the toy and its total surface area ($\pi = 3.14$).

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Worksheet – Answer Key

MATHS PART 1

Part 1 – Chapter 1 : Linear Equations in Two Variables

- Q1. (1) C (2) B [$5 \times 4 - 3 \times (-2) = 26$] (3) A [$D = 3(-1) - 2(1) = -5$] (4) B (5) B
- Q2. (1) Subtract: $2a = 4$, $a = 2$, $b = 4$. (2) $x = 3$, $y = 1$. (3) Subtract: $-4y = -4$, $y = 1$, $x = 5$. (4) $m = 6n - 7$ gives $27n = 54$; $n = 2$, $m = 5$. (5) Adding: $x + y = 5$; subtracting: $x - y = 1$; so $x = 3$, $y = 2$. (6) Put $1/x = p$, $1/y = q$: $2p + 3q = 13$, $5p - 4q = -2$; $p = 2$, $q = 3$; so $x = 1/2$, $y = 1/3$.
- Q3. (1) $D = 25$, $D_x = 50$, $D_y = -25$; $x = 2$, $y = -1$. (2) $D = -7$, $D_x = -21$, $D_y = 14$; $x = 3$, $y = -2$. (3) $D = -13$, $D_x = -52$, $D_y = -13$; $x = 4$, $y = 1$.
- Q4. (1) $x + y = 6$ passes through $(0, 6)$, $(3, 3)$, $(6, 0)$; $x - y = 2$ passes through $(0, -2)$, $(2, 0)$, $(4, 2)$. The lines meet at $(4, 2)$, so $x = 4$, $y = 2$. (2) $2x + y = 8$ passes through $(0, 8)$, $(3, 2)$, $(4, 0)$; $x - y = 1$ passes through $(0, -1)$, $(1, 0)$, $(3, 2)$. The lines meet at $(3, 2)$, so $x = 3$, $y = 2$.
- Q5. (1) 40 and 23. (2) Son 9 years, father 41 years. (3) Fixed charge ₹ 5, ₹ 10 per km. (4) The number is 36 ($x + y = 9$, $y - x = 3$; $63 - 36 = 27$).

Part 1 – Chapter 2 : Quadratic Equations

- Q1. (1) C (2) A (3) B (4) C (5) A (6) A
- Q2. (1) $2x^2 - 5x + 3 = 0$; $a = 2$, $b = -5$, $c = 3$. (2) $x^2 + x - 3 = 0$; $a = 1$, $b = 1$, $c = -3$.
- Q3. (1) $(x - 3)(x - 4) = 0$; $x = 3$ or 4 . (2) $(x + 4)(2x - 3) = 0$; $x = -4$ or $3/2$. (3) $(3m - 2)(2m - 3) = 0$; $m = 2/3$ or $3/2$.
- Q4. $x^2 + 6x + 9 = 16$, $(x + 3)^2 = 16$, $x + 3 = \pm 4$; $x = 1$ or $x = -7$.
- Q5. (1) $x = (6 \pm \sqrt{20})/2 = 3 \pm \sqrt{5}$. (2) $\Delta = 25$; $y = (7 \pm 5)/4 = 3$ or $1/2$.
- Q6. (1) (a) $\Delta = 0$, real and equal roots. (b) $\Delta = 9 - 40 = -31$, roots are not real. (c) $\Delta = 13$, real and unequal roots. (2) $k^2 - 36 = 0$, $k = 6$ or -6 .
- Q7. (1) Sum = $10/3$, product = $8/3$. (2) $x^2 + 3x - 28 = 0$. (3) $\alpha + \beta = 5$, $\alpha\beta = 6$; $\alpha^2 + \beta^2 = 25 - 12 = 13$.
- Q8. (1) $x(x + 1) = 132$, $x^2 + x - 132 = 0$, $x = 11$ ($x = -12$ rejected); integers 11 and 12. (2) $x(x + 3) = 70$, $x^2 + 3x - 70 = 0$, $x = 7$; breadth 7 m, length 10 m. (3) $x^2 + (x + 2)^2 = 130$, $x^2 + 2x - 63 = 0$, $x = 7$; numbers 7 and 9.

Part 1 – Chapter 3 : Arithmetic Progression

- Q1. (1) B (2) B (3) B (4) B (5) C
- Q2. (1) 10, 15, 20, 25 (2) -3, -5, -7, -9
- Q3. (1) $3 + 14 \times 4 = 59$. (2) $4 + (n - 1)5 = 109$, $n = 22$; the 22nd term. (3) $5 + (n - 1)6 = 301$ gives $n - 1 = 49.33$, which is not a whole number; so 301 is not a term. (4) $5d = 25$, $d = 5$, $a = 7$; $t_{15} = 7 + 14 \times 5 = 77$.
- Q4. (1) $S_{20} = 10[6 + 76] = 820$. (2) 5050. (3) 50 odd numbers; sum = 2500. (4) $n(51 - 3n) = 156$, $n^2 - 17n + 52 = 0$, $n = 4$ or 13 (both give sum 78).
- Q5. (1) $S_{12} = 6[2000 + 2200] = ₹ 25,200$. (2) $S_{15} = (15/2)[40 + 28] = 510$ chairs.

Part 1 – Chapter 4 : Financial Planning

- Q1. (1) B (2) C (3) B (4) C (5) C
- Q2. (1) GST = 12% of 8000 = ₹ 960; CGST = ₹ 480, SGST = ₹ 480; invoice value = ₹ 8960. (2) Output GST = ₹ 9000; ITC = 18% of 30,000 = ₹ 5400; GST payable = ₹ 3600; CGST = ₹ 1800, SGST = ₹ 1800.
- Q3. (1) ₹ 140. (2) $200 + 0.60 = ₹ 200.60$. (3) Brokerage per share = ₹ 0.20; $500 \times 39.80 = ₹ 19,900$. (4) Dividend per share = ₹ 15; total dividend = ₹ 3000; investment = ₹ 24,000; rate of

return = 12.5%. (5) Brokerage = ₹ 50; GST = ₹ 9 (CGST ₹ 4.50, SGST ₹ 4.50); total = 10,000 + 50 + 9 = ₹ 10,059. (6) Cost per share = 40.20; shares = 1000; brokerage = ₹ 200.

Q4. (1) NAV = ₹ 30. (2) 600 units. (3) $2000 \times 60 = ₹ 1,20,000$.

Part 1 – Chapter 5 : Probability

Q1. (1) C (2) B (3) C (4) B (5) B

Q2. (1) $S = \{HH, HT, TH, TT\}$; $n(S) = 4$. (2) $S = \{1H, 2H, 3H, 4H, 5H, 6H, 1T, 2T, 3T, 4T, 5T, 6T\}$; $n(S) = 12$.

Q3. (1) $3/6 = 1/2$ (2) $\{5, 6\}$: $2/6 = 1/3$ (3) 1 (4) 0

Q4. $n(S) = 36$. (1) (3,6), (4,5), (5,4), (6,3): $4/36 = 1/9$. (2) $6/36 = 1/6$. (3) $6/36 = 1/6$.

Q5. (1) $4/52 = 1/13$ (2) $26/52 = 1/2$ (3) $12/52 = 3/13$ (4) $1/52$ (5) $13/52 = 1/4$

Q6. $n(S) = 12$. (1) $5/12$ (2) $9/12 = 3/4$ (3) $7/12$

Q7. $n(S) = 8$. (1) $3/8$ (2) $7/8$

Q8. (1) $\{5, 10, 15, 20, 25\}$: $5/25 = 1/5$. (2) $\{2, 3, 5, 7, 11, 13, 17, 19, 23\}$: $9/25$.

Part 1 – Chapter 6 : Statistics

Q1. (1) B (2) B (3) A (4) C

Q2. Class marks 5, 15, 25, 35, 45; $\Sigma f = 50$; $\Sigma fx = 25 + 120 + 375 + 560 + 270 = 1350$; Mean = $1350/50 = 27$. Assumed mean $A = 25$: $d = -20, -10, 0, 10, 20$; $\Sigma fd = -100 - 80 + 0 + 160 + 120 = 100$; Mean = $25 + 100/50 = 27$.

Q3. Class marks 25, 35, 45, 55, 65; $u = -2, -1, 0, 1, 2$; $\Sigma f = 40$; $\Sigma fu = -8 - 8 + 0 + 10 + 8 = 2$; Mean = $45 + (2/40) \times 10 = 45.5$.

Q4. $\Sigma f = 50$; cumulative frequencies 4, 15, 35, 44, 50; $N/2 = 25$, so median class is 30 – 40. $L = 30$, $cf = 15$, $f = 20$, $h = 10$; Median = $30 + (25 - 15)/20 \times 10 = 35$.

Q5. Modal class 30 – 40; $L = 30$, $f_1 = 20$, $f_0 = 8$, $f_2 = 12$, $h = 10$; Mode = $30 + (20 - 8)/(40 - 8 - 12) \times 10 = 30 + 6 = 36$.

Q6. Food = $8000/24000 \times 360^\circ = 120^\circ$; Rent = 90° ; Education = 60° ; Transport = 30° ; Savings = 60° (total 360°).

Q7. Histogram : rectangles of width 10 on class limits 10 – 60 with heights 4, 11, 20, 9, 6.

Frequency polygon : join the midpoints (15, 4), (25, 11), (35, 20), (45, 9), (55, 6), and close the polygon at the class marks 5 and 65 (frequency 0).

MATHS PART 2

Part 2 – Chapter 1 : Similarity

Q1. (1) B (2) B (3) B (4) A

Q2. $A(\triangle ABC) = \frac{1}{2} \times 9 \times 5 = 22.5$; $A(\triangle PQR) = \frac{1}{2} \times 10 \times 6 = 30$; ratio = 3 : 4.

Q3. (1) $AD/DB = AE/EC \Rightarrow 4/6 = 3/EC \Rightarrow EC = 4.5$. (2) $AE/AC = AD/AB = 2/5 \Rightarrow AE = 4$.

Q4. $LP/PM = 3/6 = 1/2$ and $LQ/QN = 2/4 = 1/2$. Since the ratios are equal, $PQ \parallel MN$ (converse of the basic proportionality theorem).

Q5. $AB/AC = BD/DC \Rightarrow 6/9 = 4/DC \Rightarrow DC = 6$.

Q6. Ratio of areas $81 : 144 = 9 : 16$, so ratio of sides = 3 : 4. $AB/PQ = 3/4 \Rightarrow AB = 6$ cm.

Q7. (1) $AB/PQ = 4/6 = 2/3$ and $AC/PR = 6/9 = 2/3$ with $\angle A = \angle P$; similar by SAS test. (2) $3/6 = 4/8 = 5/10 = 1/2$; similar by SSS test.

Q8. $AB/BC = DE/EF \Rightarrow 6/9 = 8/EF \Rightarrow EF = 12$.

Q9. Height/6 = $20/4 \Rightarrow$ height = 30 m.

Part 2 – Chapter 2 : Pythagoras Theorem

Q1. (1) B (2) C (3) C (4) B

Q2. (1) $\sqrt{81 + 144} = 15$ cm. (2) $\sqrt{625 - 49} = 24$ cm.

Q3. (1) $49 + 576 = 625 = 25^2$; yes, right-angled. (2) $25 + 36 = 61 \neq 64$; no.

- Q4. $QS^2 = PS \times SR = 36 \Rightarrow QS = 6$. $PR = 13$; $PQ^2 = PS \times PR = 52 \Rightarrow PQ = 2\sqrt{13}$; $QR^2 = SR \times PR = 117 \Rightarrow QR = 3\sqrt{13}$.
- Q5. $AB^2 + AC^2 = 2AM^2 + 2BM^2 \Rightarrow 81 + 169 = 200 + 2BM^2 \Rightarrow BM = 5$; $BC = 10$.
- Q6. (1) $\sqrt{(100 - 64)} = 6$ m. (2) $\sqrt{(1681 - 1600)} = 9$ cm. (3) $6\sqrt{3}$ cm. (4) Half-diagonals 8 and 15; side = $\sqrt{(64 + 225)} = 17$ cm.

Part 2 – Chapter 5 : Co-ordinate Geometry

- Q1. (1) C (2) A (3) B (4) A (5) C
- Q2. (1) $\sqrt{(25 + 144)} = 13$. (2) $\sqrt{(36 + 64)} = 10$.
- Q3. $AB = 3$, $BC = 4$, $AC = 5$; $AB^2 + BC^2 = 9 + 16 = 25 = AC^2$, so $\triangle ABC$ is right-angled at B.
- Q4. Slope of $AB = (3 - 1)/(3 - 1) = 1$; slope of $BC = (5 - 3)/(5 - 3) = 1$. Equal slopes with common point B, so A, B, C are collinear. (Also $AB + BC = 2\sqrt{2} + 2\sqrt{2} = 4\sqrt{2} = AC$.)
- Q5. (1) (1, 2). (2) B = (5, 8).
- Q6. (1) $x = (2 \times 6 + 3 \times 1)/5 = 3$; $y = (2 \times (-3) + 3 \times 7)/5 = 3$; the point is (3, 3). (2) $((1 + 3 + 5)/3, (2 - 4 + 8)/3) = (3, 2)$.
- Q7. (1) $-6/6 = -1$. (2) $\tan 45^\circ = 1$; $\tan 60^\circ = \sqrt{3}$. (3) Both slopes equal 2, so the lines are parallel.
- Q8. $(x - 3)^2 + 16 = 25 \Rightarrow (x - 3)^2 = 9 \Rightarrow x = 6$ or $x = 0$.

Part 2 – Chapter 6 : Trigonometry

- Q1. (1) B (2) B (3) C (4) A (5) B
- Q2. $AC = \sqrt{(64 + 36)} = 10$ cm; $\sin A = BC/AC = 3/5$; $\cos A = AB/AC = 4/5$; $\tan A = BC/AB = 3/4$.
- Q3. (1) $1/4 + 1 = 5/4$. (2) $2 \times 1/2 + 3 \times 1/4 = 7/4$. (3) $3/4 + 1/4 = 1$.
- Q4. $\cos \theta = 12/13$; $\tan \theta = 5/12$.
- Q5. $\cos \theta = 12/13$, $\sin \theta = 5/13$, $\tan \theta = 5/12$, $\operatorname{cosec} \theta = 13/5$, $\cot \theta = 12/5$.
- Q6. (1) $\text{LHS} = \cos^2 \theta \times \sec^2 \theta = 1$. (2) $\text{LHS} = [\sin^2 \theta + (1 + \cos \theta)^2] / [\sin \theta (1 + \cos \theta)] = (2 + 2\cos \theta) / [\sin \theta (1 + \cos \theta)] = 2/\sin \theta = 2 \operatorname{cosec} \theta$. (3) $\operatorname{cosec} \theta - \sin \theta = \cos^2 \theta / \sin \theta$; $\sec \theta - \cos \theta = \sin^2 \theta / \cos \theta$; $\tan \theta + \cot \theta = 1/(\sin \theta \cos \theta)$; product = 1.
- Q7. (1) $10 \sin 60^\circ = 5\sqrt{3} = 8.65$ m. (2) $\tan 45^\circ = h/30 \Rightarrow h = 30$ m. (3) $\tan 30^\circ = 60/d \Rightarrow d = 60\sqrt{3} = 103.8$ m. (4) $h = 20 \tan 60^\circ = 20\sqrt{3} = 34.6$ m.

Part 2 – Chapter 7 : Mensuration

- Q1. (1) B (2) A (3) B (4) B
- Q2. (1) Arc = $(90/360) \times 2 \times 22/7 \times 14 = 22$ cm; area = $(1/4) \times 22/7 \times 196 = 154$ cm². (2) Arc = $(60/360) \times 2 \times 22/7 \times 21 = 22$ cm; area = $(1/6) \times 22/7 \times 441 = 231$ cm².
- Q3. Sector area = $(1/4) \times 3.14 \times 100 = 78.5$ cm²; $A(\triangle OAB) = \frac{1}{2} \times 10 \times 10 = 50$ cm²; area of minor segment = $78.5 - 50 = 28.5$ cm².
- Q4. (1) $CSA = 2\pi rh = 440$ cm²; $V = \pi r^2 h = 1540$ cm³. (2) $l = 10$ cm; $TSA = \pi r(r + l) = 3.14 \times 6 \times 16 = 301.44$ cm²; $V = (1/3) \times 3.14 \times 36 \times 8 = 301.44$ cm³. (3) $SA = 4\pi r^2 = 5544$ cm²; $V = (4/3)\pi r^3 = 38,808$ cm³. (4) $CSA = 2\pi r^2 = 308$ cm²; $TSA = 3\pi r^2 = 462$ cm²; $V = (2/3)\pi r^3 = 2156/3 \approx 718.67$ cm³.
- Q5. (1) $(4/3)\pi(6)^3 = (1/3)\pi r^2(24) \Rightarrow r^2 = 36 \Rightarrow r = 6$ cm. (2) $V = 22/7 \times 1.96 \times 3 = 18.48$ m³ = 18,480 litres. (3) Volume = $(2/3)\pi(27) + (1/3)\pi(9)(4) = 18\pi + 12\pi = 30\pi = 94.2$ cm³; slant height of cone = 5 cm; $TSA = \pi(3)(5) + 2\pi(9) = 33\pi = 103.62$ cm².